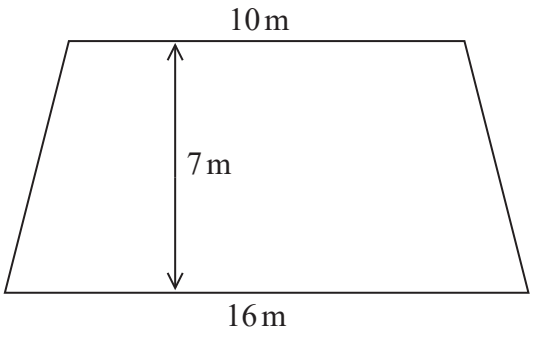
5 The diagram shows a floor in the shape of a trapezium.





£16.99

John is going to paint the floor.

Each 5 litre tin of paint costs £16.99
1 litre of paint covers an area of 2 m^2

John has £160 to spend on paint.

Has John got enough money to buy all the paint he needs?
You must show how you get your answer.

Area of trapezium = $\frac{1}{2} \times (10 + 16) \times 7 = 91\ m^2$

1 mark
Use the trapezium area

	Litres	m^2	
×45.5	1	2	×45.5
	45.5	91	

1 mark
Convert area to litres

Tins needed = 45.5 ÷ 5 = 9.1

1 mark
Find the number of tins

Buy 10 whole tins to have enough paint.

	Tins	GBP	
×10	1	16.99	×10
	10	169.90	

1 mark
Cost the rounded-up number of tins

No. £169.90 > £160, so he is £9.90 short.

1 mark
Decision supported by the full cost

(Total for Question 5 is 5 marks)